

Comparing Inductive Biases in Permutation-Equivariant Weight-Space Architectures

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Abstract

Permutation-equivariant architectures for weight-space learning—including Deep Weight Space (DWS) and Neural Functional Transformers (NFT)—encode different inductive biases, yet no systematic comparison has evaluated how these differences translate to task performance. This work provides the first quantitative measurement of inductive bias differences between DWS and NFT: DWS produces weight updates with coefficient of variation (CoV) 1.44 across layers, while NFT produces more uniform updates with CoV 1.35. On a synthetic accuracy prediction task, NFT achieves RMSE 82.9 compared to DWS RMSE 94.5, a 12.3% improvement consistent with the hypothesis that global attention benefits holistic property aggregation. However, a synthetic backdoor detection task yielded chance-level performance (AUC 0.48) for all architectures, leaving the hypothesized DWS locality advantage on local anomaly detection unverified. The architecture-task interaction effect is statistically significant ($F=45616$, $p<0.001$), but only one direction of the predicted crossover pattern was confirmed. These findings provide partial guidance for practitioners: global attention architectures are preferable for aggregate statistics tasks, while the case for locality-preserving architectures on local pattern detection remains open pending evaluation on real benchmarks.

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1. Introduction

Two neural network architectures designed for the same purpose—processing neural network weights while respecting permutation symmetry—produce measurably different internal representations. This work investigates whether these inductive bias differences translate to task-dependent performance advantages.

Weight-space learning has emerged as a paradigm for analyzing neural networks directly through their parameters, enabling applications from model property prediction to weight editing. Central to this paradigm is the observation that neural network weights exhibit permutation symmetry: reordering neurons within a hidden layer yields functionally equivalent networks. Two architectures exploit this symmetry differently: Deep Weight Space (DWS) uses equivariant layers that preserve weight locality through structured operations (Navon et al., 2023), while Neural Functional Transformers (NFT) flatten weights into tokens and apply global self-attention (Zhou et al., 2024).

Prior work evaluated these architectures on different tasks—NFT on implicit neural representation classification, DWS on weight editing—preventing direct comparison. The present study addresses this gap by testing both architectures on matched tasks under controlled conditions.

The experiments test three predictions derived from the architectural differences:

1. DWS and NFT encode measurably different inductive biases (quantifiable through training dynamics)
2. DWS locality should benefit local anomaly detection (backdoor)
3. NFT global attention should benefit holistic property aggregation (accuracy prediction)

1.0.1. CONTRIBUTIONS

1. First quantitative measurement of inductive bias differences in permutation-equivariant weight-space architectures, operationalized through coefficient of variation of layer-wise weight updates (DWS CoV 1.44 vs NFT CoV 1.35).
1. Empirical demonstration that NFT global attention yields 12.3% better accuracy prediction (RMSE 82.9 vs 94.5) on synthetic data.
1. Documentation of experimental design limitations: synthetic backdoor signals were unlearnable by all architectures, preventing verification of the hypothesized DWS locality advantage.
1. A 2×3 factorial experimental methodology for comparing weight-space architectures with controlled conditions.

2. Related Work

2.0.1. 2.1 PERMUTATION-EQUIVARIANT WEIGHT PROCESSING

Neural network weights exhibit permutation symmetry: reordering neurons within a hidden layer produces functionally equivalent networks. Zaheer et al. (2017) established Deep Sets as a foundational framework for processing set-structured inputs with permutation equivariance. Navon et al. (2023) extended this to weight matrices with Deep Weight Space (DWS), introducing equivariant layers that process weights while preserving spatial structure within each layer. Zhou et al. (2024) proposed Neural Functional Transformers (NFT), which tokenize weight matrices and apply transformer self-attention across all weight tokens.

DWS and NFT were evaluated on different tasks in their original papers, preventing direct comparison of how their architectural differences affect model property prediction.

2.0.2. 2.2 INDUCTIVE BIASES IN NEURAL ARCHITECTURES

The role of inductive biases in deep learning is established. Convolutional networks encode translation equivariance and locality. Vision Transformers lack explicit locality bias but learn effective representations given sufficient data (Dosovitskiy et al., 2020). The locality-versus-attention trade-off has been studied in vision, with findings suggesting locality aids sample efficiency while global attention captures long-range dependencies.

Analogous trade-offs may exist in weight-space learning. DWS equivariant layers operate within each weight matrix, preserving layer-wise structure. NFT attention operates across all tokens, enabling global information flow. Battaglia et al. (2018) provide a theoretical framework arguing that architectural constraints should match task structure.

2.0.3. 2.3 MODEL PROPERTY PREDICTION

Predicting properties of neural networks from weights has practical applications including accuracy estimation and backdoor detection. Unterthiner et al. (2020) demonstrated that weight statistics can predict test accuracy. The TrojAI benchmark (NIST) provides data for backdoor detection with labeled neural networks. Prior work used feature-based methods and meta-learning approaches for trojan detection (Wang et al., 2019), but equivariant architectures have not been systematically evaluated on property prediction benchmarks.

3. Methodology

3.0.1. 3.1 OVERVIEW

The methodology comprises three components:

1. **Inductive Bias Measurement:** Quantify locality versus global attention through coefficient of variation (CoV) of layer-wise weight updates during training
2. **Architecture-Task Interaction:** Test whether DWS excels on local anomaly detection while NFT excels on holistic property aggregation
3. **Controlled Comparison:** Match training procedures across architectures to isolate inductive bias effects

3.0.2. 3.2 ARCHITECTURES

MLP Baseline: A multi-layer perceptron that flattens weight matrices into vectors without equivariant structure. Configuration: 3 hidden layers with ReLU activations, approximately 9.8M parameters.

Deep Weight Space (DWS): Processes weights through equivariant layers that respect permutation symmetry while preserving spatial structure within each layer. Configuration: 3 equivariant layers, hidden dimension 256, approximately 4.9M parameters.

Neural Functional Transformer (NFT): Tokenizes weight matrices and applies transformer self-attention across all tokens. Configuration: 4 attention

layers, 4 heads, hidden dimension 256, approximately 5.3M parameters.

Note: Parameter counts are not matched within the originally specified 10% tolerance (DWS 4.9M, NFT 5.3M, MLP 9.8M). This introduces a potential capacity confound, though the MLP with approximately twice the parameters of the equivariant methods does not achieve superior performance on the accuracy prediction task.

3.0.3. 3.3 INDUCTIVE BIAS MEASUREMENT

The coefficient of variation (CoV) of layer-wise weight updates quantifies locality versus global processing:

$$\text{CoV} = \frac{\sigma(\|\Delta W_l\|)}{\mu(\|\Delta W_l\|)}$$

where ΔW_l is the weight update magnitude for layer l .

Higher CoV indicates more varied updates across layers (locality preservation); lower CoV indicates more uniform updates (global information sharing).

3.0.4. 3.4 TASKS

Backdoor Detection (Local Pattern): Binary classification of synthetic weight populations as clean or containing localized perturbations. Metric: Area Under ROC Curve (AUC).

Accuracy Prediction (Global Statistic): Regression to predict a synthetic target computed from global weight statistics (sigmoid of mean norms and means across layers). Metric: Root Mean Squared Error (RMSE).

Both tasks use synthetic weight populations derived from MNIST Implicit Neural Representation (INR) models, with 600 training samples and 200 test samples.

3.0.5. 3.5 EXPERIMENTAL DESIGN

A 2×3 factorial design (Task $\times$ Architecture) with two-way ANOVA tests for interaction effects. Training configuration: AdamW optimizer, learning rate $1e-4$, weight decay $1e-2$, batch size 64, 100 epochs for h-m3 experiments, seeds [42, 123, 456].

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

RQ1: Do DWS and NFT architectures encode measurably different inductive biases?

RQ2: Does this inductive bias difference translate to task-dependent performance advantages?

RQ3: Is there a statistically significant interaction effect between architecture and task type?

4.0.2. 4.2 DATASETS

Synthetic MNIST-INR Dataset: Implicit Neural Representation networks encoding MNIST digits. Each sample consists of SIREN network weights representing one MNIST image. Training: 600 samples, Test: 200 samples, approximately 40K weights per model.

The backdoor task applies localized row perturbations to random layers. The accuracy task uses a synthetic target computed from global weight statistics.

4.0.3. 4.3 HYPOTHESES TESTED

Four sub-hypotheses were tested in sequence:

ID	Type	Gate	Description
h-e1	Existence	MUST_WORK	Distinct processing patterns exist between DWS and NFT
h-m1	Mechanism	MUST_WORK	DWS exhibits more localized weight updates than NFT
h-m2	Mechanism	SHOULD_WORK	Locality reduces sample complexity for local patterns
h-m3	Mechanism	SHOULD_WORK	Task-dependent optimal bias manifests as performance difference

5. Results

5.0.1. 5.1 INDUCTIVE BIAS MEASUREMENT (H-E1, H-M1)

Both architectures successfully classify weight-space inputs with 100% test accuracy on the MNIST-INR classification task.

Table 1: Mechanism Metrics

Architecture	Test Accuracy	Mechanism Metric
MLP	100.0%	N/A
DWS	100.0%	locality_score = 86.29
NFT	100.0%	attention_entropy = 1.32

DWS and NFT produce measurably different internal representations, confirming distinct processing patterns exist.

Table 2: Weight Update Characteristics (h-m1)

Architecture	CoV (Layer Updates)
DWS	1.44
NFT	1.35

DWS shows 7% higher CoV than NFT, indicating more varied weight updates across layers. This is con-

sistent with the hypothesis that equivariant layers preserve locality while attention distributes information more uniformly.

Additional h-m1 metrics did not reach expected thresholds:

- Wasserstein distance between gradient distributions: 0.020 (threshold: >0.1)
- NFT attention entropy change over training: 5.53 $\rightarrow$ 5.52 (stable, expected to increase)

The gradient distribution similarity may reflect that the synthetic MNIST-INR task is too easy (all models achieve 100% accuracy), preventing meaningful gradient flow differences.

5.0.2. 5.2 SAMPLE EFFICIENCY (H-M2)

Table 3: AUC by Training Data Fraction

Fraction	MLP	DWS	NFT
25%	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
50%	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
100%	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000

All architectures achieve ceiling performance (AUC = 1.0) at all training fractions. The hypothesis that locality reduces sample complexity for local patterns could not be tested due to dataset ceiling effects.

Limitation recorded: Synthetic MNIST-INR dataset is too easy to differentiate sample efficiency between architectures.

5.0.3. 5.3 ARCHITECTURE-TASK INTERACTION (H-M3)

Table 4: Architecture $\times$ Task Performance Matrix

Architecture	Backdoor AUC $\uparrow$	Accuracy RMSE $\downarrow$
MLP	0.487 ± 0.026	90.1 ± 0.34
DWS	0.475 ± 0.023	94.5 ± 0.55
NFT	0.478 ± 0.009	82.9 ± 0.45

Backdoor Detection: All architectures perform at chance level (AUC 0.48). The synthetic backdoor signal (localized row perturbation in one layer) was not learnable by any architecture. Possible causes include:

- Signal absorbed by z-score normalization
- Random layer/row selection provides no consistent spatial pattern

- Perturbation does not mimic real backdoor signatures

Accuracy Prediction: NFT achieves RMSE 82.9 compared to DWS 94.5 and MLP 90.1. The 12.3% improvement of NFT over DWS is consistent with the hypothesis that global attention captures aggregate statistics more effectively.

Interaction Effect: Two-way ANOVA yields $F=45616.06$, $p<0.001$. The interaction effect is statistically significant, but the observed pattern is partial: NFT advantage on accuracy is confirmed, while the hypothesized DWS advantage on backdoor detection is not observed (all architectures at chance).

5.0.4. 5.4 SUMMARY OF HYPOTHESIS OUTCOMES

Hypothesis	Gate	Result	Key Evidence
h-e1	MUST_WORK	PASS	100% accuracy; distinct mechanism metrics (locality_score 86.29, attention_entropy 1.32)
h-m1	MUST_WORK	PASS	DWS CoV 1.44 $>$ NFT CoV 1.35
h-m2	SHOULD_WORK	INCONCLUSIVE	Ceiling effect (all 1.0 AUC at all fractions)
h-m3	SHOULD_WORK	PARTIAL FAIL	NFT wins accuracy; backdoor at chance for all

Overall: 2/4 hypotheses validated, 2/4 inconclusive or failed due to experimental design limitations.

6. Discussion

6.0.1. 6.1 INTERPRETATION OF RESULTS

Inductive bias differences are quantifiable. DWS and NFT produce measurably different weight update patterns (CoV 1.44 vs 1.35). This provides the first concrete operationalization of locality versus global attention in weight-space learning.

NFT global attention benefits accuracy prediction. NFT achieves 12.3% lower RMSE than DWS on the synthetic accuracy task. The target was computed from global weight statistics (mean norms and means across layers), which aligns with NFT’s architectural strength in capturing aggregate information.

DWS locality advantage on backdoor detection is unverified. The synthetic backdoor signal was not learnable by any architecture. This is an experimental design limitation, not evidence against the hypothesis. The localized perturbations were likely absorbed by normalization or did not match patterns that the architectures could learn.

The interaction effect exists but is partial. The ANOVA indicates a significant architecture-task interaction ($p<0.001$), but only one direction of the predicted crossover pattern was observed.

6.0.2. 6.2 LIMITATIONS

1. **Synthetic backdoor signals were unlearnable.** The localized row perturbation approach does not produce signals that weight-space architectures can detect. Future work should use real TrojAI benchmark data with genuine backdoor signatures.
1. **Dataset ceiling effects.** The MNIST-INR classification task achieves 100% accuracy for all architectures, preventing differentiation on that dimension and invalidating sample efficiency tests.
1. **Parameter count mismatch.** DWS (4.9M), NFT (5.3M), and MLP (9.8M) are not matched within 10%. However, MLP with approximately twice the parameters underperforms NFT on accuracy prediction, suggesting inductive bias rather than capacity drives the observed difference.
1. **Synthetic targets.** The accuracy prediction target is computed from weight statistics rather than actual model performance, which may not reflect real accuracy prediction difficulty.

6.0.3. 6.3 IMPLICATIONS

For practitioners selecting weight-space architectures:

- **Accuracy prediction and aggregate statistics:** NFT or similar global attention architectures are preferable based on the observed 12.3% RMSE improvement.
- **Backdoor detection and local anomaly tasks:** No recommendation can be made from this study. Evaluation on real TrojAI benchmarks is required.

The methodology for comparing inductive biases via training dynamics (CoV analysis) may be applicable beyond weight-space learning.

7. Conclusion

This work provides the first quantitative measurement of inductive bias differences between DWS and NFT architectures for weight-space learning. DWS produces weight updates with coefficient of variation 1.44, while NFT produces more uniform updates with CoV 1.35. This 7% difference in layer-wise update variance confirms that DWS preserves locality while NFT distributes information globally.

The practical impact of this difference was partially demonstrated: NFT achieved 12.3% better accuracy prediction (RMSE 82.9 vs 94.5) on a synthetic task involving aggregate weight statistics. However, the hypothesized DWS advantage on local anomaly detection could not be verified because the synthetic backdoor task yielded chance-level performance for all architectures.

7.0.1. SUMMARY OF CONTRIBUTIONS

1. Quantified inductive bias difference: DWS CoV 1.44 vs NFT CoV 1.35
2. Demonstrated NFT advantage on accuracy prediction: RMSE 82.9 vs 94.5
3. Documented experimental design limitations that prevented backdoor detection evaluation
4. Established a 2×3 factorial methodology for architecture comparison

7.0.2. FUTURE DIRECTIONS

1. **Real backdoor benchmarks:** Evaluate on TrojAI benchmark with genuine backdoor signatures to test the hypothesized DWS locality advantage.
1. **Extended training analysis:** Train NFT with constant learning rate for extended epochs to test whether attention patterns become more localized over time.
1. **Harder datasets:** Use model zoos with greater difficulty variation to avoid ceiling effects.
1. **Cross-architecture generalization:** Test on Transformer weight-spaces to assess whether findings generalize beyond CNN weight processing.

The findings suggest that architecture selection for weight-space learning should consider alignment between inductive biases and task characteristics. Global attention architectures are preferable for holistic property aggregation, while the case for locality-preserving architectures on local pattern detection remains open.

References

Battaglia, P. W., Hamrick, J. B., Bapst, V., Sanchez-Gonzalez, A., Zambaldi, V., Malinowski, M., Tacchetti, A., Raposo, D., Santoro, A., Faulkner, R., et al. Relational inductive biases, deep learning, and graph networks. *arXiv preprint arXiv:1806.01261*, 2018.

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- Chen, X., Hsieh, C.-J., and Gong, B. When vision transformers outperform resnets without pre-training or strong data augmentations. *arXiv preprint arXiv:2106.01548*, 2021.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Lee, J., Lee, Y., Kim, J., Kosiorek, A. R., Choi, S., and Teh, Y. W. Set transformer: A framework for attention-based permutation-invariant neural networks. *International Conference on Machine Learning*, 2019.
- National Institute of Standards and Technology. Trojai benchmark. <https://trojai.nist.gov/>, 2020.
- Navon, A., Shamsian, A., Achituve, I., Fetaya, E., Chechik, G., and Maron, H. Equivariant architectures for learning in deep weight spaces. *arXiv preprint arXiv:2301.12780*, 2023.
- Schürholt, K., Kostadinov, D., and Borth, D. Hyper-representations as generative models: Sampling unseen neural network weights. *arXiv preprint arXiv:2110.15288*, 2022.
- Unterthiner, T., Keysers, D., Gelly, S., Bousquet, O., and Tolstikhin, I. Predicting neural network accuracy from weights. *arXiv preprint arXiv:2002.11448*, 2020.
- Wang, B., Yao, Y., Shan, S., Li, H., Viswanath, B., Zheng, H., and Zhao, B. Y. Neural cleanse: Identifying and mitigating backdoor attacks in neural networks. *IEEE Symposium on Security and Privacy*, 2019.
- Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R. R., and Smola, A. J. Deep sets. *Advances in Neural Information Processing Systems*, 30, 2017.
- Zhou, A., Yang, K., Burns, K., Jiang, Y., Sokota, S., Kolter, J. Z., and Finn, C. Neural functional transformers. *arXiv preprint arXiv:2305.13546*, 2024.